Introduction

Meta-analysis of diagnostic test accuracy pools per-study sensitivity and specificity to summarise how well a test discriminates disease from non-disease across multiple primary studies. Unlike meta-analysis of treatment effects, the two paired outcomes are intrinsically correlated: sensitivity and specificity are determined jointly by the test threshold, so a study choosing a stricter cut-off will trade higher specificity for lower sensitivity. Naively pooling each metric separately ignores this trade-off and produces estimates that misrepresent the test's underlying performance. The modern approach is bivariate random-effects modelling (Reitsma et al., 2005) or hierarchical summary ROC modelling (Rutter and Gatsonis, 2001), both of which respect the correlation structure and yield interpretable summary points and curves.

Prerequisites

A working understanding of sensitivity, specificity, and the receiver-operating characteristic (ROC) curve; familiarity with the logit transformation; and basic mixed-effects modelling concepts.

Theory

The bivariate random-effects model jointly fits logit-sensitivity and logit-specificity using a bivariate Normal random effect across studies, estimating pooled sensitivity, pooled specificity, between-study variances for each, and the correlation between them. The HSROC model parameterises the underlying ROC curve directly with random study-specific thresholds and accuracy parameters, producing a summary curve that explicitly accommodates threshold variation. Reitsma's bivariate model and the HSROC model are mathematically equivalent under standard parameterisation; the choice between them is mainly one of presentation — pooled point with confidence ellipse versus summary curve.

Assumptions

Each study contributes a 2×2 table of true positives, false negatives, false positives, and true negatives. Random effects on the logit scale are bivariate Normal; threshold variation across studies is captured implicitly by the random-effects structure; case ascertainment and reference-standard verification are comparable across studies (otherwise applicability concerns enter, captured by the QUADAS-2 framework).

R Implementation

```r
library(mada)

set.seed(2026)
df <- data.frame(
  TP = c(45, 60, 32, 80, 28, 55),
  FN = c(5, 10, 8, 20, 12, 5),
  FP = c(10, 15, 8, 20, 10, 12),
  TN = c(140, 165, 152, 180, 150, 138)
)

fit <- reitsma(df)
summary(fit)

plot(fit, sroclty = 1, sroclwd = 2,
     main = "Summary ROC curve")
```

Output & Results

`reitsma()` returns pooled logit-sensitivity and logit-specificity with their variances and the between-study correlation; back-transformation gives the pooled point estimates with confidence intervals and a confidence ellipse on the ROC plane. The summary ROC curve traces the implied trade-off across thresholds, and a summary AUC quantifies discrimination across the full operating range.

Interpretation

A reporting sentence: "Bivariate meta-analysis of six diagnostic-accuracy studies (Reitsma model, REML) pooled sensitivity 0.84 (95 % CI 0.78 to 0.89) and specificity 0.91 (95 % CI 0.87 to 0.94) on the logit scale; the summary ROC area under the curve was 0.94. The between-study correlation −0.3 indicates threshold variation contributes meaningfully to between-study heterogeneity, which the bivariate model accommodates explicitly. Reporting follows PRISMA-DTA." Always report both summary points and the SROC curve, plus the between-study correlation.

Practical Tips

- Always use bivariate or HSROC models; separate univariate pooling of sensitivity and specificity is statistically incorrect because it ignores the threshold-induced correlation between the two.
- Report the pooled point estimate, the 95 % confidence ellipse on the ROC plane, and the summary curve — together they convey both central performance and uncertainty.
- Threshold variability is a major source of heterogeneity in DTA meta-analyses; subgroup analyses by pre-specified threshold or by clinical setting are often more interpretable than a single overall pool.
- For studies with zero counts in any cell, `mada` applies a 0.5 continuity correction by default; sparse-data settings may benefit from Bayesian DTA-NMA models in `bamdit` or `INLA`.
- PRISMA-DTA is the reporting standard for diagnostic-accuracy systematic reviews; QUADAS-2 is the standard risk-of-bias assessment tool for individual primary studies.
- For diagnostic-accuracy NMA — comparing multiple competing tests on the same disease — `BUGSnet`, `bamdit`, and recent extensions in `multinma` provide the appropriate framework.

R Packages Used

`mada::reitsma()` for the canonical bivariate random-effects model and `mada` for SROC plotting and AUC; `mada::HSROC()` for the equivalent HSROC parameterisation; `bamdit` for Bayesian DTA meta-analysis with rich diagnostics; `meta::metaprop()` for univariate proportion pooling when only one of sensitivity or specificity is of interest; `multinma` for Bayesian DTA-NMA across multiple competing tests.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Systematic reviews and PRISMA
- Meta-analysis basics
- Network meta-analysis